GPS, GALILEO, and Nanotechnology: A Cost-Effective Satellite Technology?

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***Abstract*--The prospects of cost-effective Earth observation missions, using the global positioning system (GPS) and Galileo satellites (GIOVE-A and GIOVE-B) and associated telemetry designs, are migrating toward economical satellites, operational ground stations, efficient data distribution structure, and qualitative space system management.**

The concept of application specific integrated micro-instruments (ASIM) —microelectronics, integration of micro-electromechanical systems (MEMS), miniaturization, and signal conditioning—enables the development of small satellite technologies. However, cost-effective management of nano- and pico-satellites, rapid data distribution methods, and quality assurance procedures of these mission developments, are far from achievable standard and performance capacity.

Thus, this study presents the status quo, possibilities and prospects of small satellite mission activities, research budget and cost drivers toward attaining cost-effective missions, with visions of the future missions and greater involvement of local and small-scale industry.

I. THE REAL RACE IN GNSS: GPS AND GALILEO AT A GLANCE

The evolution of the Global Navigation Satellite System (GNSS) into the 21st century has witnessed at least 4 global systems and at least 6 regional systems in Europe, the America, and Asia. These evolutions, a giant leap of emerging technology, are aimed at fostering new GNSS applications, timely and sufficient satellite availability issues, downstream services to end users, and manufacturer benefits—European Union (EU) or non-EU industries. The obvious effects on GNSS developments and policies, as Galileo tend to Full Operational Capability (FOC) in 2014 [1] with a new era of GPS III satellites modernization, reflects in the implementation of additional budget funding of advanced space vehicles (SVs). Table 1 is a summary of the active GNSS constellation status, featuring related satellite based augmentation systems (SBAS) for accuracy, reliability, and availability.

Galileo, a joint initiative of the European Commission (EC) and the European Space Agency (ESA), is still in the development phase. Under the auspices of the European GNSS Supervisory Authority (GSA), it assumed Galileo management on January 1, 2007. Galileo In-Orbit Validation Element-A (GIOVE-A) was launched successfully into space, in December 28, 2005, the first of the two Galileo satellites that is to secure the system's frequency allocation and validate key technologies for the full Galileo constellation. GIOVE-A, on carrier frequencies E1 (1191.795MHz), E5 (1278.750MHz), and E6 (1575.420MHz), continues to transmit prototype test signals [2] on E1/E5 and E1/E6 band combinations. The GIOVE-B signal recorded its first measurements during the night of May 7, 2008, following the earlier successful lunch from Baikonur on April 26, 2008, and still continues its test transmissions. The successful completion of the In-Orbit Test (IOT) phase has been marked by one year signal quality assessment by European Space Agency (ESA) GIOVE mission activities to stabilize the ground control segment [3], [4].

Following an initial launch failure in 1986, twelve of the Block IIR satellites were launched between 1997 and 2004 to replenish the GPS constellation as older satellites were decommissioned. The GPS constellation has experienced an additional L5 (1176.45MHz) signal to its "legacy signals", the L1 (1575.42MHz) and L2 (1227.60MHz) signals, from the recently launched IIR (M)-20 satellite (SVN49), on March 24, 2009 [5]. This new signal, compared with the current L1 and L2 frequencies, is intended for safety-of-life (SoL) applications, and is envisaged to improve, potentially, ionospheric refraction corrections[6], [7]—this translate invariably to better quality measurements— and GPS receiver tracking performance. The concept of L2C, or L2 civil signal [8], under GPS Modernization program, was to circumvent the need of dual-frequency operations for direct cancellation of ionospheric biases, using the modernized Block IIR and Block IIF satellites. Presently, six Block IIR-M satellites are part of the 35 current GPS satellite constellation. Past GNSS developments, augmentation services, and future constellation developments are shown in a timeline in Figure 1.

However, the GNSS evolutions have not been without its problems [9], [10]. The SVN49, as a demonstration payload for the Block IIF and Block III satellite still in early development stages, is not yet in full compliance with the stipulated IS-GPS-705 GPS specifications. The L5 signal, unsuitable for navigation, contains only data-less Q5 component modulated with the pseudorandom noise code PRN63. This was due, in part, to the rush to secure the L5 GPS frequency allocation before the August 2009 deadline by International Telecommunication Union Radiocommunication Sector (ITU-R), without the first planned L5-capable space vehicle launch. Further, the GPS constellation with a *de facto* limit of 31 satellites was designed as a 24-satellites constellation. To date, there are 35 satellites in the constellation with 30 satellites set 'healthy' to users. These aging GPS satellites, availability gaps, glitches, and vulnerabilities from delayed constellation replenishment are without concerns in the GNSS community.

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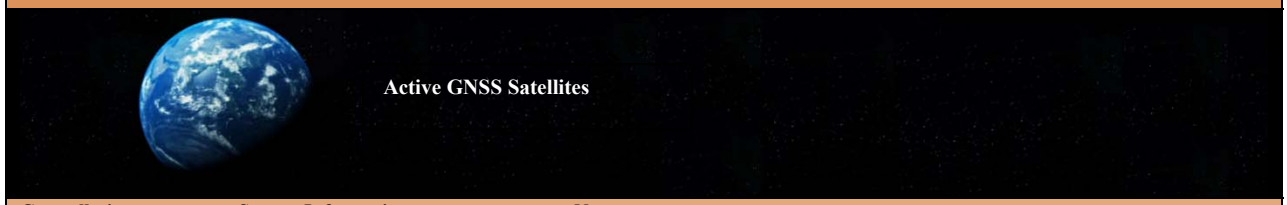
TABLE 1. THE ACTIVE GNSS CONSTELLATION STATUS, AUGMENTATION SYSTEMS, AND SATELLITE INFORMATION.



Constellations	System Information	Notes
GPS	11 Block IIA 12 Block IIR 7 Block IIR-M	A total of 30 usable satellites under FOC (Full Operational Capability). Selective Availability (SA) was set to zero on 2nd May 2000, 4:00 UT Antispoofing (AS) was activated on 31st January, 1994, on all Block IIs. April 10, 2009, GPS L5 signal transmission began.
Galileo	GIOVE-A GIOVE-B	Transmit two signal combinations: E1+E5 and E1+E6 GIOVE-A operates on rubidium atomic clock, and GIOVE-B operates on hydrogen maser atomic clock.
GLONASS	18 GLONASS-M	The first GLONASS satellite was launched 12th October, 1982. All use cesium atomic clock
Beidou/Compass	Beidou 1A (GEO) Beidou 1B (GEO) Beidou 1C(GEO) Beidou 1D(GEO) Beidou 1M (MEO) Beidou G2 (GEO)	Beidou 1D is currently adrift Beidou G2 is now at 74ºE and is drifting
Satellite Based Augmentation Systems (SBAS)		
SBAS	**Satellite (PRN)**	**Notes**
EGNOS	Inmarsat-3-F2/AOR-E(120) Artemis(124) Inmarsat-3-F5/IOR-W(126)	EGNOS started Open Service on 1st October, 2009
GAGAN	Inmarsat-4-F1/IOR(127)	
MSAS	MTSAT-1R (129) MTSAT-2 (137)	MSAS was commissioned for aviation use on 27th September , 2007
WAAS	Intelsat Galaxy XV (135) TeleSat Anik F1R (138)	

Should the focus on the rapid expansion of space system management be redirected toward cost-effective missions, using emerging technologies? Could an attractive option of the small satellite initiative booster the needed space access? What impact would it have on scheduled mission cost and durations? These outstanding issues necessitate an integrate review of prospective Earth Observation mission activities and programs.

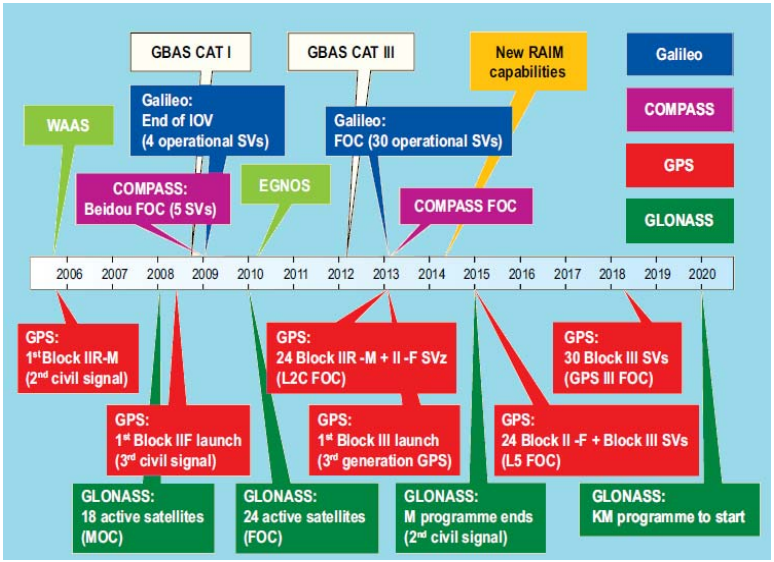


Fig.1. The developments of GNSS constellations and global augmentation systems (Source:www.gpsworld.com)

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II. EARTH OBSERVATION MISSIONS: ORGANIZATIONS AND PROGRAMS

Sustained improvements of the international collaborations in Earth observation missions, using operational structures, are effected by the establishment and operations of working groups and task groups. These workgroups and task groups, implicitly, are responsible for the exchange of policies and technical information, facilitation of services and development of Earth observation activities, and implementation of compatibility among spaceborne Earth observation systems. Table 2 shows a summary of major organizations and programs involved in Earth observation activities used for research and scientific applications.

III. CEOS

The Committee on Earth Observation Satellite (CEOS), established in 1984 in response to a recommendation from a Panel of Experts on Remote Sensing from Space, and under the aegis of the G7 Economic Summit of Industrialized Nations Working Group on Growth, Technology, and Employment, is coordinating the multidisplinary and international civil spaceborne missions of satellite Earth observations. CEOS comprise of its Members, Associates, and the Secretariat. The Secretariat coordinate and maintain the plenary sessions by the European Space Agency (ESA) and the European Organization for the Exploitation of Meteorological Satellites (EUMETSAT), the National Aeronautics and Space Administration (NASA) and the National Oceanic and Atmosphere Administration (NOAA) of U.S.A, and the Ministry of Education, Culture, Sports, Science and Technology (MEXT) jointly with the Japan Aerospace Exploration Agency (JAXA). Up to date, 28 space agencies along with 20 other national and international organizations continue to actively participate in CEOS planning and activities. Globally, the committees present the most comprehensive, and up-to-date technology and services status related to governmental, space-based Earth observation programmes. Successful completion, on December 4, of ESA's Soil Moisture and Ocean Salinity (SMOS) mission—Launch and Early Orbit Phase (LEOP)—is included in the 2009 achievements of the CEOS [11].

IV. ESA

Galileo is a vision and global ambition of the European Commission (EC) and ESA, for advanced GNSS, providing highly accurate and robust global positioning services under civilian control. ESA on 15, June, 2009, as part of the Galileo FOC procurement programme, signed two contracts for procurement of Long-Lead Items required for the construction of the Galileo satellite constellation. ESA, established in 1975, is jointly responsible with EC for the management of Galileo definition and development phase, and in-orbit validation (IOV) phase [12]. The purpose of IOV phase is to mitigate risks associated with proto-flight model—verification of interfaces, electronic and electrical functionalities, and validation of payload performances. It has scheduled first 2 IOV satellite launch in November 2010, with a targeted initial constellation of 16 satellites—four IOV and 12 FOC satellite—"operational early 2014" [13].

TABLE 2.THE GNSS ORGANIZATIONS, SUPPORT PROGRAMS, AND INTERNATIONAL COOPERATION.

Notable Earth Observation Organizations

Organizations	Subgroups/Programs	Functions
United Nations (UN)	UN committee on Peaceful Uses of Outer Space (UN/COPOUS) UN Environment Programme- Global Resource Information Database (UNEP-GRID) UN Institute for Training and Research (UNITAR) Operational Satellite Applications Programme(UNOSAT) UN Office for Outer Space Affairs (UNOOSA)	Conference activities, seminar and training programs in areas of disaster management, land cover monitoring, metrological satellite uses, cartography, and environmental monitoring.
European Commission (EC)	European Geostationary Navigation Overlay Service (EGNOS) Galileo	Open and Commercial Service, Safety-of-Life (SoL), Transportation
Committee on Earth Observation Satellites (CEOS)	Strategic Implementation Team (SIT) Working Group on Information Systems and Services (WGISS) Working Group on Calibration and Validation (WGCV) Working Group on Education and Training (WG-EDU)	Facilitates collaboration among international users' community and its members, international coordination of space-related Earth observation activities, and exchange of policy and technical information.
National Aeronautics and Space Administration (NASA)	NASA Marshall Space Flight Center (NASA-MSFC) NASA Jet Propulsion Laboratory (NASA-JPL) NASA Goddard Space Flight Center (NASA-GSFC) NASA Ames (NASA AMES)	To pioneer the future in space exploration, scientific discovery and aeronautics research.
European Space Agency (ESA)	ESA European Space Research Institute (ESA-ESRIN) ESA European Space Research and Technology Centre (ESTEC) ESA European Space Operations Centre (ESA-ESOC) ESA European Astronaut Centre (ESA-EAC) ESA European Space Astronomy Centre (ESA-ESAC) Guiana Space Centre (CSG)	Space Science Missions, Earth Observation Missions, and nurtures scientific and industrial competence in its member states in global environmental issues, telecommunications and aerospace industries.
Europe's Metrological Satellite Organization (EUMETSAT)	EUMETCast Open Geospatial Consortium (OGC) Initiative GEONETCast SIMDat Heterogeneous Mission Accessibility (HMA) Initiative	To establish, maintain and exploit European systems of operational meteorological satellites for monitoring climate and detection of global climate changes.

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Competing offer of two consortia, the EADS Astrium-ThalesAleniaSpace partnership and OHB-Surrey Satellite Technology Limited (SSTL) conglomerate, to build out the constellation, reached a peak on January 7, 2010 when, the European Commission announced the award of contracts for deployment of Galileo's initial operational capability in space. The first order of 14 satellites goes to OHB System AG of Bremen, Germany, while a contract for system support services is bestowed upon ThalesAleniaSpace of Italy, and one for launch services to Arianespace of France. The remaining three procurement contracts, for the ground mission infrastructure, the ground control infrastructure, and the operations is scheduled to be awarded middle of 2010 [14], [15], [16].

V. NASA

For nearly 50 years, NASA has pioneered space exploration, scientific discovery, and aeronautic research. NASA developed Ground Network (GN)—a global array of ground stations—Deep Space Network (DSN), and Space Network (SN)—a network of geosynchronous communications satellites—for communications and navigation [17]. The synergy of planned NASA networks and GPS Terrestrial Service Volume service, an ongoing collaboration for outer space navigation, will create an international environment to support, challenge the interoperability of the existing global systems, and further energizing the 2004 U.S. Space Exploration Policy. Present GPS integrations, the Miniaturized Airborne GPS Receiver (MAGR-S) on the receiver Space Shuttles and the Space Integrated GPS/INS (SIGI) receiver on the International Space Station (ISS), highlight certain successes in this direction. By 2011, as part of NASA's vision for space, GPS telemetry data would be transmitted down to mission controllers, using GPS Metric Tracking (GPS-MT) services, complementing radar network positioning. These integrated networks are directed toward future space architecture, optimized operations and network capabilities, and cost-effective Earth observation missions [18].

VI. USAF GPSW

The Global Positioning System Wing (GPSW) is a service effort directed by U.S Air Force (USAF), and managed at the Space and Missile Systems Center (SMC) unit, Los Angeles Air Force Base, California. The primary responsibility to track and maintain command control, alert warning, update navigational, and communications satellites, for Air Force Space Command (ASFPC), is carried out by the 50th Space Wing (50 SW). An incisive review of the GPS III satellite program, and established, reliable baseline review with regard to the cost, schedule and technical requirements, with Lockheed Martin team, projects the launch schedule of the first two GPS IIIA satellites for 2014 [19]. GPS III constellation would feature cross-linked command, control architecture, and simultaneous update of satellites from a single ground station.

Although characterized with major achievements, these featured organizations, however, are yet to address in recent years, the challenges of maintenance of operational satellites and additional cost accrued due to delayed schedules. The anticipated future capabilities and endangered constellation stability, and current technical problems threaten signal availability to end-users and the revolutionized GNSS industries. Perhaps, the emerging small satellite technologies, as key to cost reduction, improvement of satellite synthesis and fabrication, and optimization of technology management, could contribute significantly to space segment re-design? Innovation in technology requires substantial performance and economic benefit.

VII. NANOTECHNOLOGY: A SMALL SATELLITE INITIATIVE

"Nano", the current scientific innovation catchphrase, has featured enormous economic benefit and competitive landscape in the 21st century. The sustainable innovations, technological growth, and international collaborations in nanoscience, for various fields of science and technology, are making significant impacts on research and development [20]-[24]. The evolution of nano centers, all over the world, has addressed technology-related issues —focusing on overcoming scientific limits, and developing sound business models [25].

A major advancement of nanotechnology—fabrication of advanced materials at the atomic level— is in the area of micro-electronic mechanical systems (MEMS). Assessment of the global MEMS device market survey forecast potential rise to $95 billion (Figure 2), a 13% compound annual growth rate (CAGR) for this year alone—with a trailing record of opportunities, challenges, and investment trends of 15% compound annual growth rate for the past five years. This has accentuated the clamor for application specific integrated micro-instruments (ASIM), especially in front-end manufacturing, which accounts for 53% MEMS equipment market. On October 13, 2008, TriQuint Semiconductors unveiled its integrated 3 × 3 millimeters RF front-end modules for GPS–based navigation systems, the smallest package on the market [26]. This model's filter, designed for wireless application, provide rejection superior to other devices on the market. Infineon Technologies GPS receiver front-end module released two days after, measures 2.5 × 2.5 × 2.5 millimeters and is based on silicon germanium (Si-Ge) process technology [27]. The rapid market growth advances toward the main challenge—achieving higher sensitivity, and higher immunity against interferences of cellular signals. Over 200 million of these packages are in use by the global GPS manufactures from these 2 companies alone.

The rising tides of miniaturization in microelectronics—nanowires, nanotubes, nanofibres, and integrated solid state circuits—are not lost on Galileo and

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GPS industries [28]. Intelligent technology, arising from development of smart materials, biosensors, and miniature GPS/Galileo positioning modules has pushed, to an unprecedented scale, the power management efficiency, boosted signal acquisition and performances (Wi-Fi, Bluetooth), and lowered the cost of future seamless GPS/Galileo consumer application integrations. The culminating effect, of these manufacturing capabilities, is the transformation of research and development landscape in Earth observation mission to smaller cost-effective missions, with emphasis on small satellite technology [29]-[31].

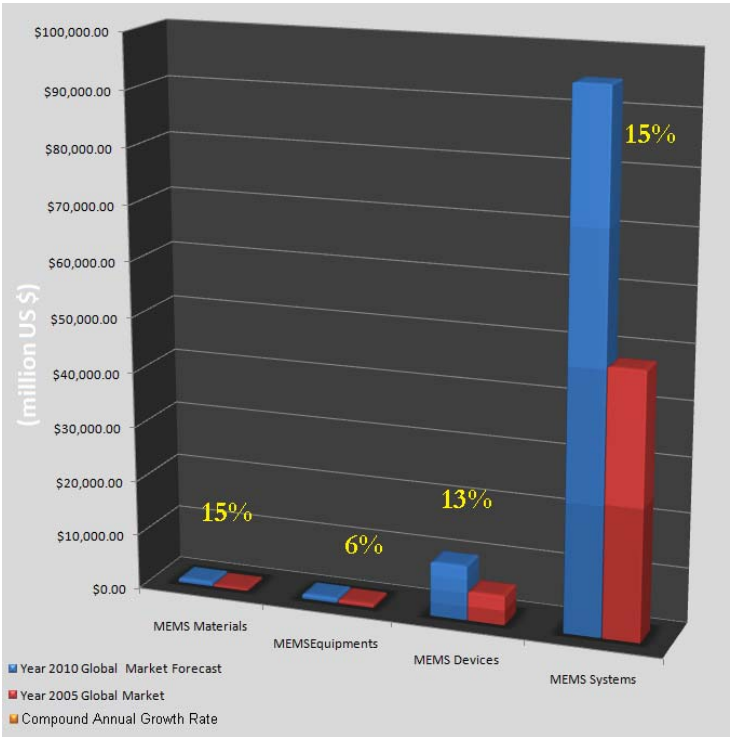

Fig. 2. The worldwide MEMS market and forecasts (Source: Yole Développement)

Small satellite initiative, an optimized and low-cost realization of extensive Earth observation mission design using advanced technology, have made advancements, since the inception of space science, in compact sensors, small satellite bus, and launch vehicle technology management. The challenges of payload fabrication, manufacturing duration barriers, data resolutions and performance capabilities, are critical issues addressed as governmental agencies, commercial entities, and organizations are facing international competition and tighter financial budget [32]-[35]. Moreover, the ability to predict the mission successes of these satellite missions is crucial to the achievement of objectives and control of bigger missions [36]. NASA, in cooperation with scientist and engineers, has developed mini-, micro-, pico-, and nano-satellites (1-1000 kg) to test innovative science and emerging engineering technologies, on a smaller scale in space and in relatively short time frames[37], [38]. On May 19, 2009, NASA lunched PharmaSat nano-satellite and completed successfully 96 hours biological experiment within six days. Further, GeneSat-1 mission, a collaboration of NASA Ames Research Center, the industries and local universities developed a fully-automated, miniaturized space flight system which provides life support and nutrient delivery, and perform assays for genetic changes in space. Moreover, IRIS nano-satellite, to be developed by Santa Clara University's Robotic Systems Laboratory as part of a comprehensive educational program, for autonomous multi-system responsive operations and high spectral resolution of Earth imaging, is an ongoing research [39], [40].

Whether these implementations are extensible to the present global systems—to a large degree—remain unanswered. Perhaps that step, in the right direction, started with chip-scale atomic clock unveiled by National Institute of Standards and Technology (NIST) and Defense Advanced Research Agency (DARPA) on August 24, 2004 —fabricated and encapsulated on semiconductor wafer, the size of about $1.5 \times 1.5 \times 4$ millimeters, it consumes less than 75×10^{-3} Watt, and loses/gains 1s for the next 300 years. Similar, but advanced prototypes, the hydrogen maser atomic clock operates in GIOVE-B. Yet, the achievements of small satellite technology indentify, to the scientific and decision-making agencies, the usefulness of nanotechnology as key to the future in GNSS architectural requirements. What identify the strategies and technologies that provide strongest benefit to both Earth and Space programs, and provide the best

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returns on investments? What are the cost-effective factors attributed to small satellite technologies? The answer lies in the market opportunities and utilization successes of the space-based satellite system.

VIII. MICRO-, PICO-, AND NANO-SATELLITES: COST ESTIMATION AND MODELING

Nanoscience has woven the technological fabric of satellite technology into a new platform approach—modularity versus integration—initiating development of small payload structures, appropriate for small satellites. Thus, weight and size constraints of present payload structures are giving way to emerging subsets of micro-, pico-, and nano-satellites. Although not a universally accepted definition, small satellites are generic terms for spacecrafts under 1000kg weight limit (Table 3), with nano- and pico-satellites under 10kg and 1kg respectively. The cost of developing typical nano-satellites, on the average, is well under US$1million and for micro-satellites between US$2-5million [41]. However, the optimal cost-effectiveness of these budget analyses for payload structures and small satellite implementation depend on mission design, mission operation, development approach, not to mention the efficient technical management and team integration. Significant emphasis on shorter mission turnaround, risk reduction of multiple platform usage, wider exploration of space in shorter time frame, and utilization of personnel at lower cost, have lent credence to the cost-effectiveness of small satellite technology. Granted, small integrated payload structures suffer limitations—limited platform resources, on-board calibrations, and thermo-elastic distortions. Still, scientifically feasible and economical viable small satellite missions are the way to the future space explorations, and next-generation nanotechnology solutions are the navigating vehicles over the demanding and challenging terrain. Table 4 summarizes major advances to payload structures and platform in small satellite implementation.

TABLE 3. A SIMPLIFIED SMALL SATELLITE NOMENCLATURE

Small Satellites	Weight
Mini-satellite	Less than 1000 kg
Micro-satellite	Less than 100kg
Nano-satellite	Less than 10 kg
Pico-satellite	Less than 1 kg

TABLE 4. MAJOR PAYLOAD STRUCTURE AND PLATFORM ADVANCES

Payloads	Platforms
Detectors	GPS receivers
Low Power electronics	Autonomous miniature star trackers
Field-Programmable Gate Arrays (FPGAs)	Control Moment Gyro (CMGs) solutions
Solid-State Memories	Battery technology
Advanced Materials	Downlink speeds

The cost estimation models of small satellite, a dependence on several factors, are estimations of the various phases and elements of the space missions. The factors include (1) the lifecycle phases, (2) the technology development phase, (3) research, development, test and evaluation, (4) the production phase, and (5) operation and maintenance. Moreover, the overall mission cost drivers place emphasis on the payload development and stability, the spacecraft reliability, quality assurance with acceptable level of risk, the ground segment—especially mission control, Earth observation data reception, distribution and archiving and operational maintenance. And, cost-effective earth observation missions can only be efficient with these operational priorities and utilizations. The best practice from micro-satellite experience, and clear trends manifested in emerging market trends of MEMS industry are identified as optimizing factors for future cost-effective Earth Observation mission. Fig. 3 show the representative cost estimation of

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typical small satellite space mission, featuring various mission cost drivers ratios. It is observed from the representative data that actual cost (with an inflation factor 1.12) for small satellite missions, compared to traditional satellite mission implementations, are dramatic in cost reduction. The tremendous cost reduction of these satellite programs (50-90%) clearly suggest the actual cost of micro-satellite projects is far below the expected cost (Figure 4).

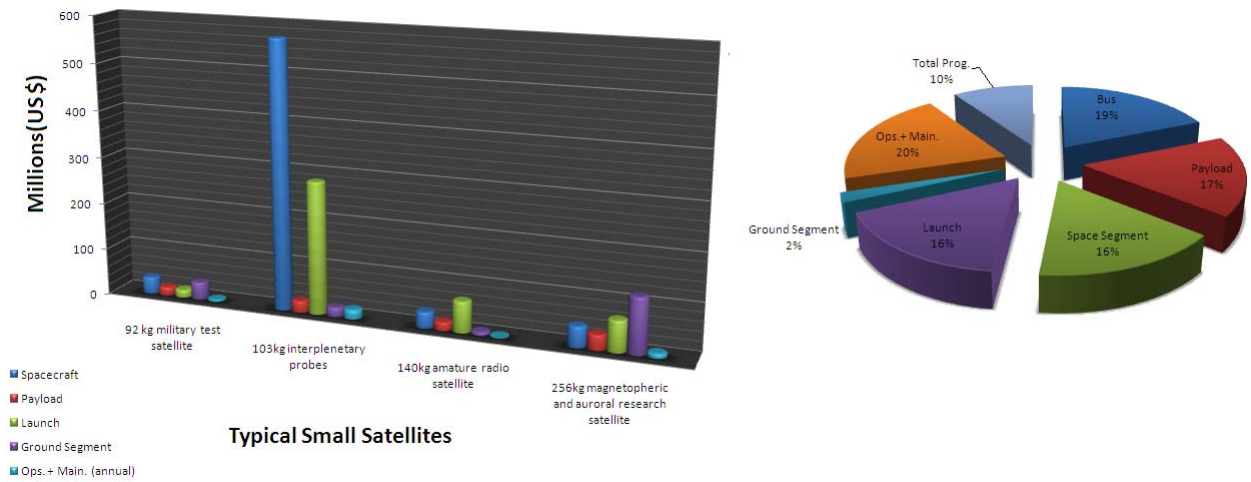

Fig. 3. Representative cost data for 10 small satellite space missions and the operational cost ratio model (Source: 3rd IAA International Symposium on Small Satellites for Earth Observation).

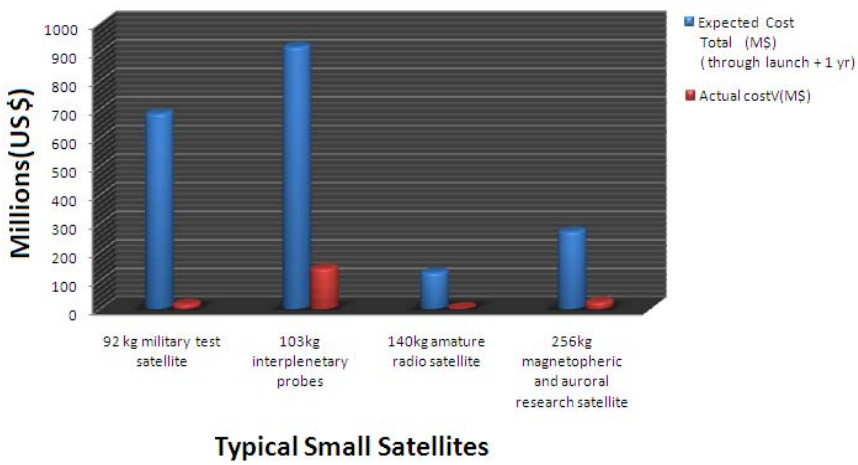

Fig. 4.The expected and actual cost data of typical small satellite space missions (Source: 3rd IAA International Symposium on Small Satellites for Earth Observation).

Certainly, the low-cost options of small satellite technology as economically viable concept to GNSS missions are key issues for consideration. These issues should be defined within the boundaries of the space mission analysis and design process—the objectives, the concept, and the evaluation of the mission. How this affects *modus operandi* of the constellation design, the mean-mission duration and the design life of these satellites are seen as trade-offs and compromises between achievable objectives and budget constraints.

IX. SMALL SATELLITE TECHNOLOGY AND MANAGEMENT: A COST-EFFECTIVE TECHNOLOGY?

There has been this ever going debate about access to space and the cost-effectiveness of the technology associated with it. The hypothetical debate, within the GNSS community, is: given the present situation of the global systems, would the GNSS community gain more from new signals, or from more satellites? What bearing and implication would this have on the associated and inevitable high cost of space programs? Is cost reduction always achievable? A key point to unraveling the mystery is a function of the perceived mission risk at both the engineering

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and management level. An equally important factor to cost-effectiveness is the cost of failure of a mission.

In April 1990, NASA initiated the possibilities of cosmic explorations— a distance ten billion light years (9.46×10^{12} km) away—through the eye of Hubble Space Telescope (HST). The $1.6 billion (US) telescope, orbiting at 500 km above the earth, was equipped with two 2.4m reflecting mirror, accurate to within two millionths of an inch. The deficient pictures sent back, a huge disappointment, indicated an unexpected major optical flaw despite the use of off-the-shelf equipments, technical expertise, and management recommendations. What actually happened? Apart from delayed HST launch and the overhead cost of keeping the team together till mission completion, the scheduled final tests were abandoned due to time limits and over budget. The cost of the ambitious repair mission, in November 1993, to fix the lens was $263 million (US) plus 30 hours of astronaut's spacewalking, to replace obsolete equipments [42].

No organization sets out to design and implement space mission without cost-effective mission analysis. However, with trends of recurrent over budget, re-evaluation of mission objectives and requirements by sponsoring organizations, to find a reasonable comprise between more expensive traditional design process and the emerging technology of design-to-cost process—trading on requirements to achieve a standard, low-cost space mission—seem necessary. Moreover, with the opt for more satellites for space exploration, high risk in mission performance over the years have been reduced with wafer-scale ASIMs, Extremely Large Scale Integration (ELSI) applications, appearance of micro launchers on the market, and multi-lateral cooperation. This has enabled the concept of small satellite technology and management. And, increase in associated performance capabilities of small satellite missions had been accredited to the management and organizational approach. A key characteristic of the small satellite management is the highly integrated small teams (of about 25 persons) working in close proximity, with good communication protocols, led by informed, and responsive management—the best structure for producing small satellites within financial considerations, performance specifications, and delivery targets [43].

Figure 5 show cost ratios of low-cost, commercially available launchers options and associated payload weight on the international market. Although the usage of small satellites for Earth observation, and as part of GNSS constellation, is limited compared to traditional space vehicles. Nonetheless, cost-effective Earth observation missions are only efficient with implementation of small satellite management concept, evident from the small launcher characteristic low-cost option. Thus, most notable successes in mission performances are directly proportionate to the integrated approach, and the development of interface standards of the mission—seen only at micro- and nano-satellite levels—enabling module compatibility and interchangeability. Further efforts and international collaborations, in the use of small satellite systems, should still be explored for the benefit of mankind.

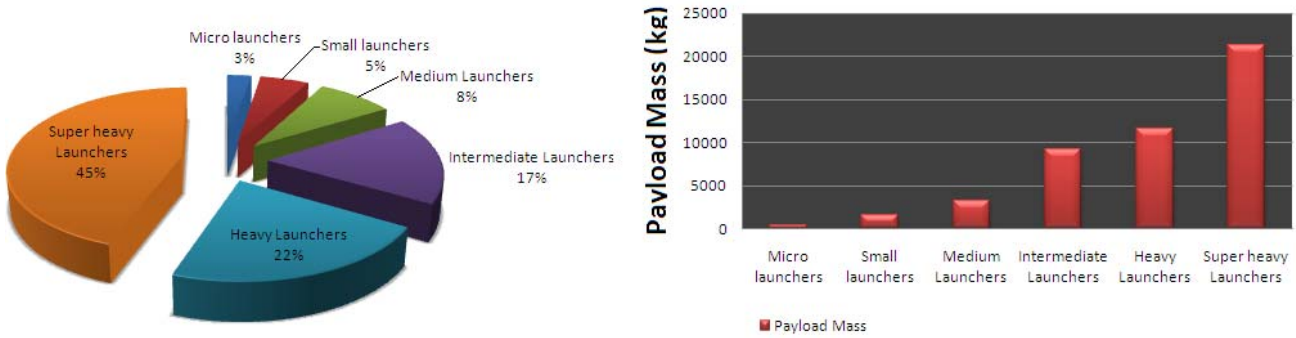

Fig. 5.The cost ratio data of international market launchers and associated payload mass (Source: 3rd IAA International Symposium on Small Satellites for Earth Observation).

X. ACHIEVING COST-EFFECTIVE MISSIONS: A GLOBAL PROBLEM, A GLOBAL SOLUTION

The explosion of technology, in the 21st century, is perhaps most apparent with proliferations of 65 satellites in earth orbits already, and a promise of increase to 90 satellite in the nearest future. The communication and navigation

revolutions, worldwide, had benefited immensely from advanced SV models (modernization) as they broadcast new signals of high integrity (E2, E6A, E6B, E5, L1, L5), with increased precise position accuracy (4 meters rms, 7.8 meters-95%) and timing [44]-[48]. However, milestones in the evolution of satellite technology is without it downsides. As newer receivers tumble in to the market to keep up with recent advents, downstream industries are catalyzing at unprecedented speed leaving critical issues unresolved. For instance, by December 31, 2020, civil use of P(Y) would no longer be supported, and today, the receivers' market-driven tendency is from codeless/semi-codeless technology with the projected economic benefit of $3.6-$4.8 billion to be lost worldwide [49]. The optimistic and realistic growth in the next multi-constellation receiver, however, initiates tougher customer requirements, perhaps, backward compatibility, and trade-off managements—a global customer support

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programme—amongst GNSS providers. Moreover, a clear perspective of the GNSS market picture is: the emerging GNSS Market will not wait for new solutions to existing problems. The tendency towards a global trade policy and disputes should not be overlooked either. And, manufacturer-friendly Interface Control documents (ICDs) remains the only the key to improving evolutions of conflicting policies. EU satellite Navigation programs for Galileo circumvented some these policy issues when, on October 2008, it announced a rescind on its requirements for a special license to manufacture and sell Galileo receivers.

"Expandable 24", or 24+3 configuration is a current GPS solution for the over populated constellation—some satellites orbiting in tandems. How this improves the standard positioning service (SPS) performance, and resolves the odds of current GPS Health issues are still in question. Based on the most recent satellite reliability and launch schedule data, the estimated long-term probability of maintaining a constellation of at least 24 operational satellites fall below 95% during fiscal year 2010 and remains below 95% until the end of fiscal year 2014[50].The worst-case scenario is to restructure the new GPS IIIA program to avoid past management flaws—schedule delays and overrun of cost estimates.

Cost of research and development has driven consolidation amongst GNSS global players. Training in technological innovations and Education outreach in Earth Observation mission, as mean of planning, analyzing, and implementing cost-effective mission successes along with it successes and failures, should be given further incentives. Presently, ESA commit 1% of its annual budget into educational outreach and activities to give awareness and insight into its space programs. NASA, for all its space missions, invests 2-3% of the mission budget in education and NASA-sponsored outreach, as initiatives and opportunities for the GNSS community. Activities, should, however, be driven toward the global small satellite community, with reviews and incorporated studies of small satellite mission promoted. These bring to light challenges and capabilities of small satellite management, compared to current knowledge, and ways of achieving low-cost small satellite mission.

The expected final transition date of all major GNSS developments is year 2020. The accelerating convergence of GNSS multi-constellation from overall cost-effective satellite replenishment, to performance of GNSS signal's interoperability, integrity, and compatibility is just around-the-corner development. As the events unfold, the economic benefits, future challenges, and new capabilities clearly stipulates small satellite technology as a huge potential to reinforcing existing GNSS missions, maximizing cost reductions, and harmonizing space mission concepts.

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